5. STABILITY

The three requirements that enter in to the design of a control system are transient response, steady state error, and stability. Of the three, stability is the most important system specification. If a system is unstable, the transient response and steady state errors are of no consequence.

A linear-time-invarient system is stable if every bounded input yields a bounded output.

The total respsonse of a system = Natural response + forced response

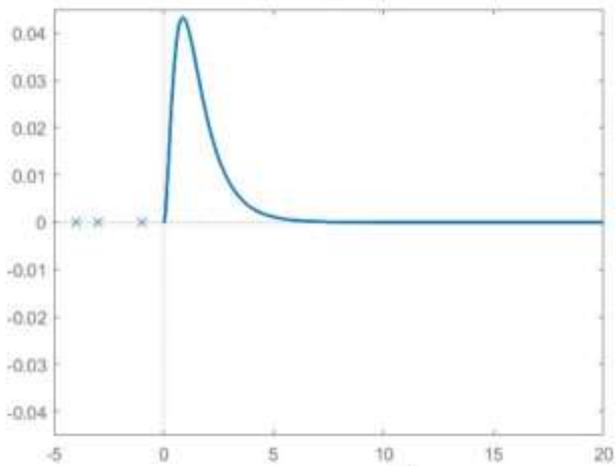
- A LTI system is stable if the natural response $\rightarrow 0$ when time $\rightarrow \infty$.
- A LTI system is unstable if the natural response grows unbounded when time $\rightarrow \infty$.
- A LTI system is marginally stable if the natural response neither decays nor grows but remain constant or oscillates as time $\rightarrow \infty$.

If all the closed loop poles of a system are located on the left half of the s-plane, then there will be a negative real part for the poles. Then the natural response will have an exponential decay term $e^{-\zeta\omega_n t}$. Hence the system will be stable.

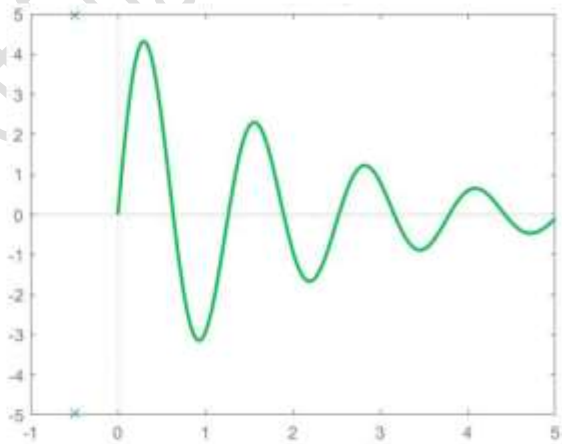
If any ofthe closed loop poles are on the right half s-plane, then the system will be unstable.Also poles of multiplicity greater than one on the imaginary axis lead to the sum of responses $At^n \sin(\omega_n t + \varphi)$, where n=1,2,3……… which also approaches infinity as time approaches infinity.

A system that has imaginary axis poles of multiplicity one yields pure sinusoidal oscillations as natural response and are considered as marginally stable.

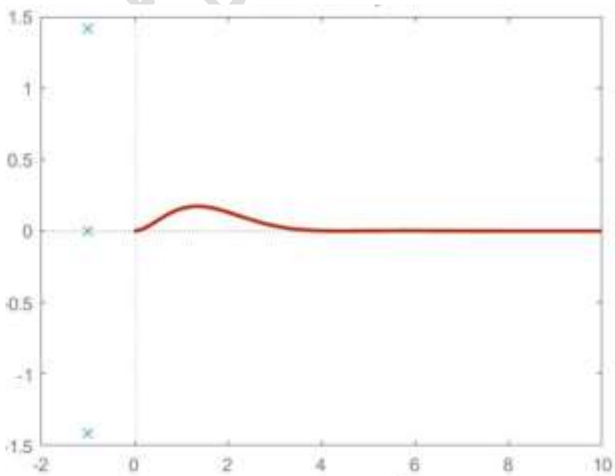
$G(s) = \frac{1}{s^3+8s^2+19s+12}$ $\quad p_i = -1, -3, -4$



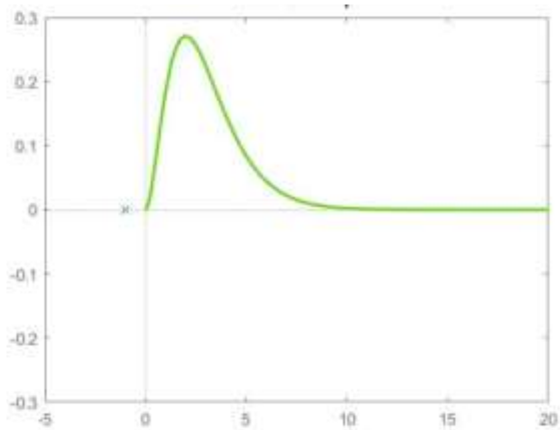
$G(s) = \frac{25}{s^2+s+25}$ $\quad p_i = -0.5 \pm j4.98$



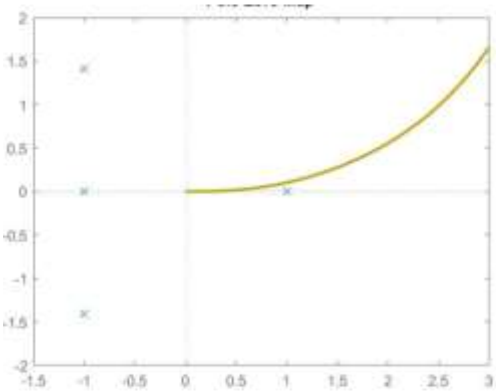
$G(s) = \frac{1}{s^3+3s^2+5s+3}$ $\quad p_i = -1, -1 \pm j\sqrt{2}$



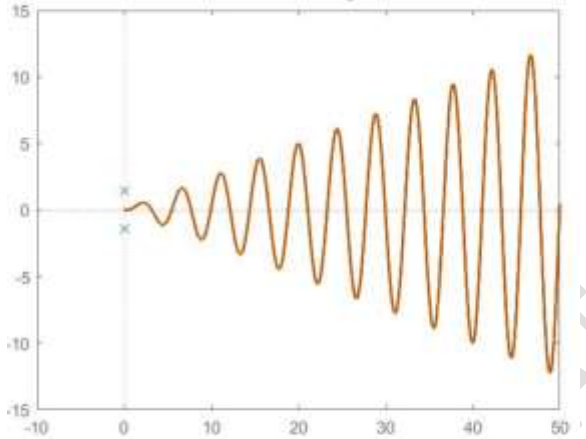
$G(s) = \frac{1}{s^3+3s^2+3s+1}$ $\quad p_i = -1, -1, -1$



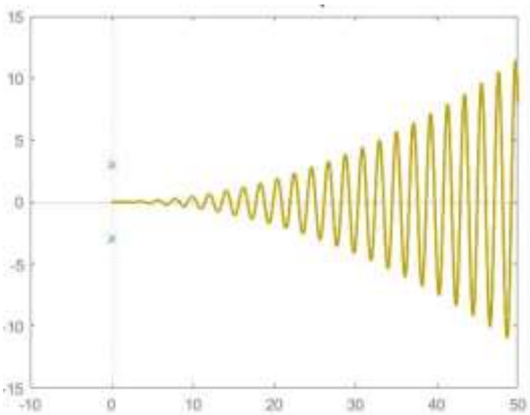
$G(s) = \frac{1}{s^4+2s^3+2s^2-2s-3}$ $p_i = -1, +1, -1 \pm j\sqrt{2}$



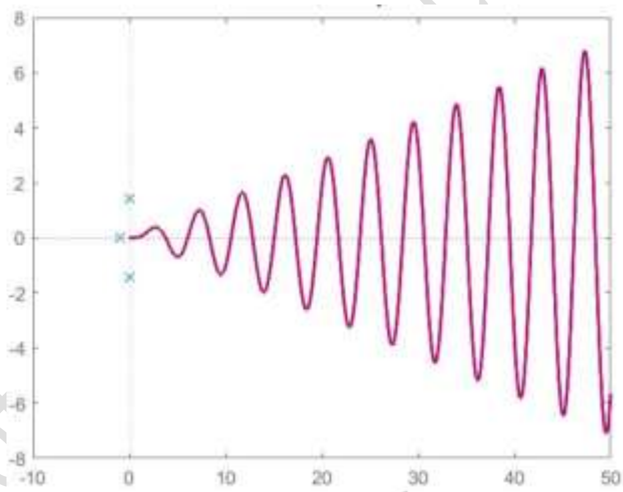
$G(s) = \frac{1}{s^4+4s^2+4}$ $p_i = \pm j\sqrt{2}, \pm j\sqrt{2}$



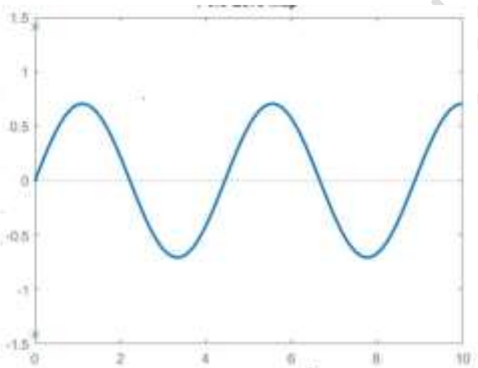
$G(s) = \frac{1}{s^6+27s^4+243s^2+729}$ $p_i = \pm j3, \pm j3, \pm j3$



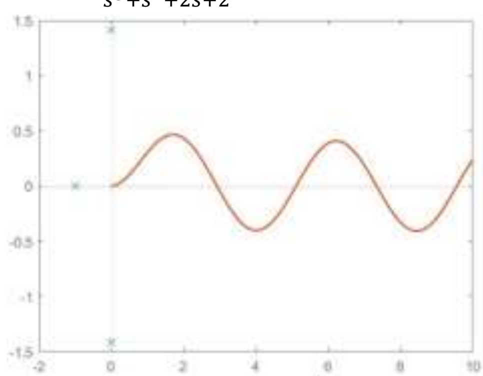
$G(s) = \frac{1}{s^5+s^4+4s^3+4s^2+4s+4}$ $p_i = -1, \pm j\sqrt{2}, \pm j\sqrt{2}$



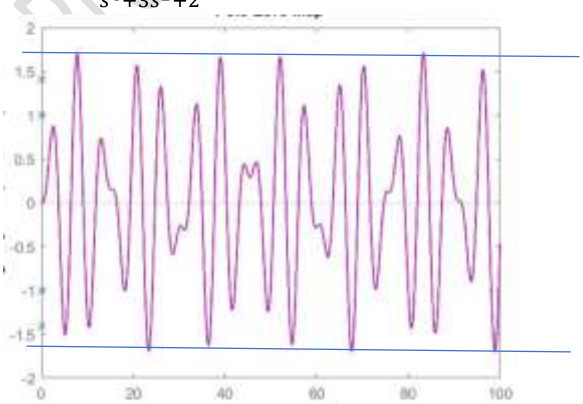
$G(s) = \frac{1}{s^2+2}$ $p_i = \pm j\sqrt{2}$



$G(s) = \frac{1}{s^3+s^2+2s+2}$ $p_i = -1, \ \pm j\sqrt{2}$



$G(s) = \frac{1}{s^4+3s^2+2}$ $p_i = \pm j\sqrt{2}, \pm j1$



Routh-Hurwitz's Stability Criterion

The control system is stable if and only if all closed loop poles lie on the left half of the s-plane. Most linear closed loop systems have their closed loop transfer function of the form

$$\frac{C(s)}{R(s)} = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

Where a's and b's are constants, and $m \le n$. Routh's stability criterion helps us to find the number of closed loop poles that lie in the right half s-plane without having to factorise the denominator polynomial.

- Write the denominator polynomial in s in the form $a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$ where the coefficients are real quantities and $a_0 \neq 0$; that is, possibility of any zero root is eliminated.
- If any of the coefficients are zero or negative, there is a root or roots that are imaginary or that have positive real parts. In such a case, the system is not stable. This is a necessary condition, but not sufficient.
- If all coefficients are positive, arrange the coefficients of the polynomial in rows and columns in the following pattern:

s^n	a_n	a_{n-2}	a_{n-4}	a_{n-6}	… … … …
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	a_{n-7}	… …
s^{n-2}	c_1	c_2	c_3	c_4	…………..
s^{n-3}	d_1	d_2	d_3	d_4	…………
……….	………	…………..	……….		…………
s^2	e_1	e_2			
s^1	f_1				
s^0	g_1				

where $c_1 = \frac{a_{n-1}.a_{n-2} - a_n.a_{n-3}}{a_{n-1}}$ $\quad c_2 = \frac{a_{n-1}.a_{n-4} - a_n.a_{n-5}}{a_{n-1}}$ $\quad c_3 = \frac{a_{n-1}.a_{n-6} - a_n.a_{n-7}}{a_{n-1}}$

$d_1 = \frac{c_1.a_{n-3} - a_{n-1}.c_2}{c_1}$ $\quad d_2 = \frac{c_1.a_{n-5} - a_{n-1}.c_3}{c_1}$ $\quad d_3 = \frac{c_1.a_{n-7} - a_{n-1}.c_4}{c_1}$

This process is continued until the last row is completed. The complete array is triangular.

The Routh's stability criterion states that the number of roots with positive real parts is equal to the number of changes in sign of the coefficients of the first column of the Routh array.

Let us take an example. Consider the transfer function $G(s) = \frac{1}{s^4+2s^3+3s^2+4s+5}$. The characteristic equation is $s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$. All the coefficents are present and all coefficients are positive. Hence, the necessary conditions for the stability of the system are satisfied. Now we can complete the Routh array.

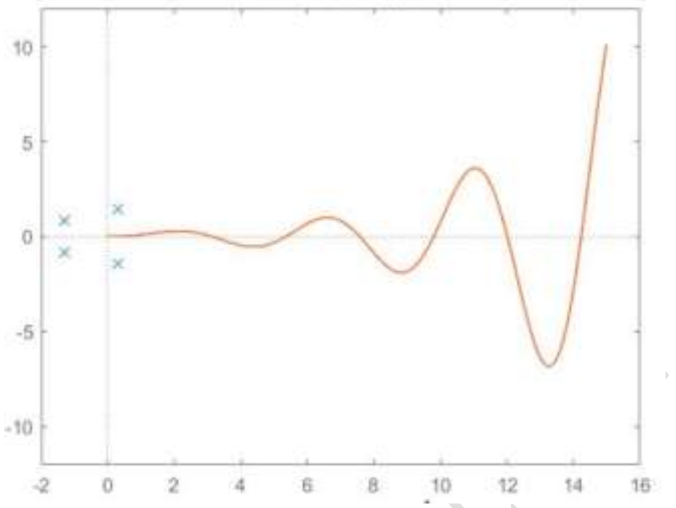
s^4	1	3	5
s^3	2	4	
s^2	1	5	
s^1	-6		
s^0	5		

$$\frac{2\times 3 - 1\times 4}{2} = 1 \qquad \frac{2\times 5 - 1\times 0}{2} = 5$$

$$\frac{1\times 4 - 2\times 5}{1} = -6$$

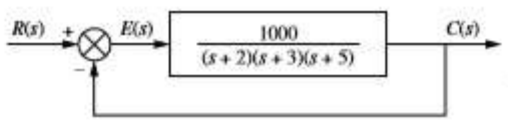
$$\frac{-6\times 5 - 1\times 0}{-6} = 5$$

There are two sign changes in the first column of the Routh's array. Hence, two roots lie in the right half s-plane. The values of the poles can be obtained using Matlab as -1.2878 + 0.8579i, -1.2878 - 0.8579i, 0.2878 + 1.4161i, and 0.2878 - 1.4161i. We can see that two roots lie on the right half s-plane. Hence the sysytem is unstable.



*Example

Using Routh-Hurwitz criterion, determine whether the system shown is stable or not.



The closed-loop trnafer function of the system is $G(s) = \frac{G}{1+GH} = \frac{1000}{s^3+10s^2+31s+1030}$

The characteristic equation is $s^3 + 10s^2 + 31s + 1030 = 0$. All coeffients are present and all are positive. The necessary conditions for stability are satisfied.

s^3	1	31	
s^2	10	1030	
s^1	-72		
s^0	103		

$$\frac{10 \times 31 - 1 \times 1030}{10} = -72$$

$$\frac{-72 \times 1030 - 10 \times 0}{-72} = 103$$

There are two sign changes in the first column of the Routh array. Therefore, two roots will lie in the right half s-plane. Hence the system is unstable.

SPECIAL CASES

1. Zero in the first column of a row

   If the first element of a row is zero, then a division by zero is necessary to form the next row. To avoid this, we replace zero with a very small quantity ε and proceed. Finally, we let $\varepsilon \to 0$ and then see the sign changes in the first column.

*Example

Check the stability of the system whose closed loop transfer function is given by $G(s) = \frac{10}{s^5+2s^4+3s^3+6s^2+5s+3}$

The characteristic equation is $s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3 = 0$. All the coefficients are present and all coefficients are positive. Hence, the necessary conditions for stability are satisfied.

Let us complete the Routh array to find how many roots are in the right half s-plane.

s^5	1	3	5
s^4	2	6	3
s^3	$\cancel{0}\ \varepsilon$	3.5	
s^2	$\frac{6\varepsilon-7}{\varepsilon}$	3	
s^1	$\frac{6\varepsilon-24.5-3\varepsilon^2}{6\varepsilon-7}$		
s^0	3		

$$\frac{2\times 3-1\times 6}{2}=0$$

$$\frac{2\times 5-1\times 3}{2}=3.5$$

$$\frac{3\varepsilon-2\times 0}{\varepsilon}=3$$

$$\frac{6\varepsilon-2\times 3.5}{\varepsilon}$$

$$\frac{\frac{6\varepsilon-7}{\varepsilon}\times 3.5-3\varepsilon}{\frac{6\varepsilon-7}{\varepsilon}}=\frac{6\varepsilon-24.5-3\varepsilon^2}{6\varepsilon-7}$$

$$\frac{\frac{6\varepsilon-24.5-3\varepsilon^2}{6\varepsilon-7}\times 3-\frac{6\varepsilon-7}{\varepsilon}\times 0}{\frac{6\varepsilon-24.5-3\varepsilon^2}{6\varepsilon-7}}=3$$

As $\varepsilon \to 0^+$, $\frac{6\varepsilon-7}{\varepsilon}$ →a negative quantity, $\frac{6\varepsilon-24.5-3\varepsilon^2}{6\varepsilon-7}$ → a positive quantity. If we look at the sign changes, there are two sign changes in the first column. Hence, two roots will lie in the right half s-plane. The system is unstable.

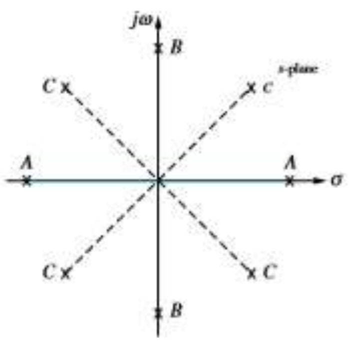
	$\varepsilon \to 0^+$	$\varepsilon \to 0^-$
s^5	+	+
s^4	+	+
s^3	+	-
s^2	-	+
s^1	+	+
s^0	+	+

2. Zeros in the entire a row

When there is an entire row of zeros, there is an even polynomial which is factor of the original polynomial. Even polynomial will have only even powers of s. They will only have roots that are symmetrical about the origin. This symmetry can occur under three conditions of root position.

1. The roots are symmetrical and real
2. The roots are symmetrical and imaginary
3. The roots are quadrantal

The row previous to the row of zeros contains the even polynomial which is a factor of the original polynomial. Everything from the row containing the even polynomial down to the end of the Routh table is a test ofonly the even polynomial



When we have a row of zeros, form the auxiliary polynomial [P(s)] using the row just above that. Determine $\frac{d}{ds}P(s)$. Replace the row of zeros by the coefficients of $\frac{d}{ds}P(s)$ and proceed as usual. Let us take an example to demonstrate this.

*Example

Check the stability of the system whose closed loop transfer function is given by $G(s) = \frac{10}{s^5+7s^4+6s^3+42s^2+8s+56}$

Construct the Routh'sarray

s^5	1	6	8
s^4	7	42	56
s^3	0	0	
s^2			
s^1			
s^0			

$$\frac{7\times 6 - 1\times 42}{7} = 0$$

$$\frac{7\times 8 - 1\times 56}{7} = 0$$

Now we have a row of zeros in the s^3 row. Form the auxiliary polynomial with the coefficients of the s^4 row.

$P(s) = 7s^4 + 42s^2 + 56$

$\frac{d}{ds}P(s) = 28s^3 + 84s$. Replace the row of zeros with these coefficients

s^5	1	6	8
s^4	7	42	56
s^3	28	84	
s^2	21	56	
s^1	9.3		
s^0	56		

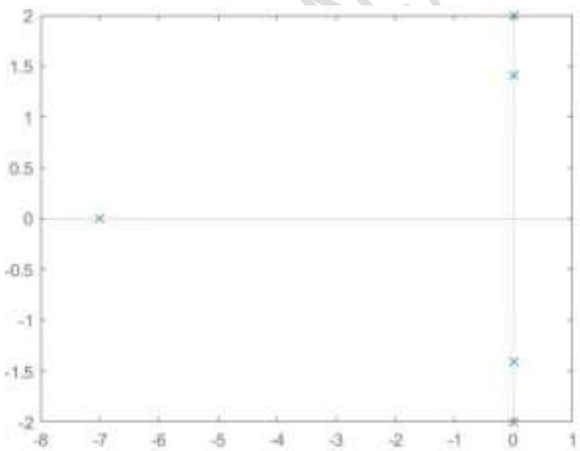
$$\frac{28\times 42 - 7\times 84}{28} = 21$$

$$\frac{28\times 56 - 7\times 0}{28} = 56$$

$$\frac{21\times 84 - 28\times 56}{21} = 9.3$$

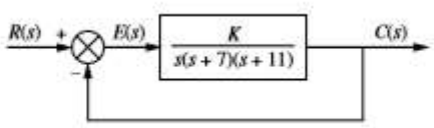
$$\frac{9.3\times 56 - 21\times 0}{9.3} = 56$$

There is no sign change in the first column of the Routh array. Hence, no roots on the right half plane. Since we have roots on the imaginary axis, the system is marginally stable. The poles are $s = -7, \pm j\sqrt{2}, \pm j2$



*Example

Find the range of K>0, for which the given system is stable, unstable, and marginally stable.



Given that the open loop transfer function $G(s)=\frac{K}{s(s+7)(s+11)}$, feedback transfer function $H(s)=1$. The closed-loop transfer function of the system is given by

$$T(s)=\frac{C(s)}{R(s)}=\frac{G}{1+GH}=\frac{\frac{K}{s(s+7)(s+11)}}{1+\frac{K}{s(s+7)(s+11)}}=\frac{K}{s^3+18s^2+77s+K}$$

Let us construct the Routh Table.

s^3	1	77
s^2	18	K
s^1	$\frac{1386-K}{18}$	
s^0	K	

s^3	1	77
s^2	18	1386
s^1	36	
s^0	1386	

For stable system, There shouldnot be a sign change in the first column. For $1386>K>0$, all the three roots lie on the LHP.

If $K>1386$ we will have a negative sign in the first column of s^1 row. Then there will be two sign changes(from +ve to -ve to +ve) in the first column. Hence two roots will lie on the RHP plane. The system will be unstable.

If K=1386, we will have a row of zeros in the s^1 row.The auxiliary polynomial is

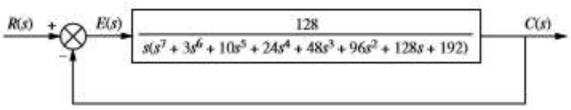
$P(s)=18s^2+1386$ $\quad\quad$ $\frac{d}{ds}P(s)=36s$

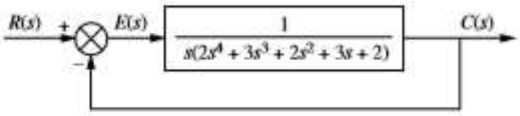
The Routh table changes to the one as shown on the left.

Again no sign changes.So, no roots on RHP. Therefore, the symmetric roots of multiplicity one lie on the imaginary axis.

#Exercise 34

Determine the number of poles on the LHP, RHP, and jw axis for the three cases shown





$$T(s)=\frac{20}{s^8+s^7+12s^6+22s^5+39s^4+59s^3+48s^2+38s+20}$$

#Exercise 35

A feedback control system has a characteristic equation $s^3+(1+K)s^2+10s+(5+15K)=0$

The parameter K has to be positive. What is the maximum value K can assume before the system becomes unstable? When K equals the maximum value, the system oscillates. Determine the frequency of oscillation.

References

1. Modern Control Systems : Richard C. Dorf and Robert H. Bishop, 8/e, Addison-Wesley
2. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE
3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons